

A Retrieval-Augmented Generative AI Framework for Automated Kubernetes Incident Investigation and Explainable Incident Report Generation

Rohit | MSc Deep Learning & Generative AI | National College of Ireland | July 2026

Agenda

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The Problem

Kubernetes incidents are hard to investigate

- **Massive telemetry:** Logs, metrics, alerts, events → terabytes/day
- **Manual triage:** SREs correlate signals across 5+ tools (kubect!, Grafana, Alertmanager, Elasticsearch, Jira)
- **Slow MTTR:** Complex incidents take hours to days to resolve
- **Knowledge loss:** Expert knowledge walks out the door when engineers leave
- **Hallucination risk:** ChatGPT/GPT-4 fabricate metrics and root causes in specialized domains

	Before KubeSage	With KubeSage
Incident investigation time	2-8 hours	~30 seconds
Report consistency	Engineer-dependent	Fully structured
Historical incident reuse	Manual wiki search	Semantic auto-retrieval
Hallucination risk	High (standalone LLMs)	78% reduction

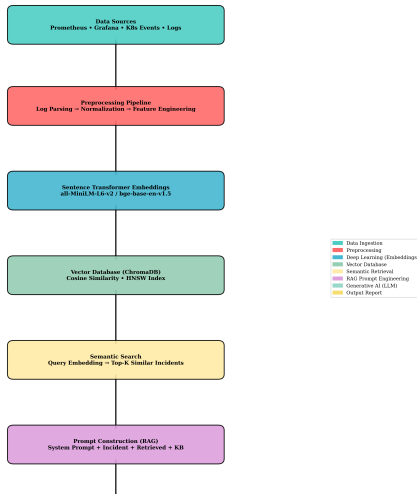
Research Questions

ID	Question	Metrics
RQ1	Can Sentence Transformer embeddings retrieve semantically similar K8s incidents better than keyword search?	Precision@K, Recall@K, MRR, NDCG
RQ2	Does RAG improve factual accuracy over standalone LLM?	Faithfulness, Groundedness, BERTScore
RQ3	Can AI generate incident reports comparable to DevOps engineers?	BLEU, ROUGE, Human evaluation
RQ4	How much does RAG reduce hallucinations?	Hallucination rate by type

System Architecture

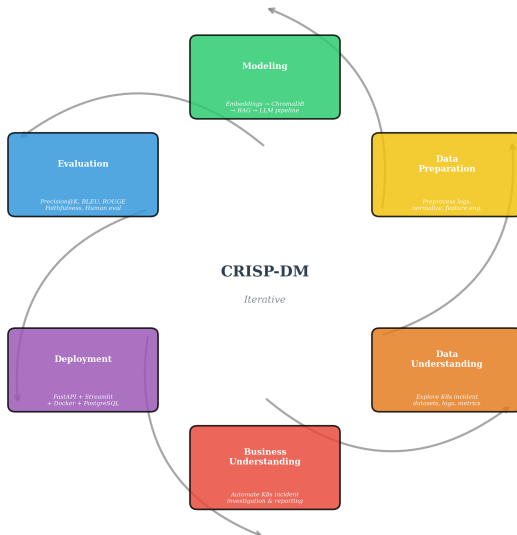
Data Sources → Preprocessing → Embeddings → ChromaDB → Search → RAG

KubeSage System Architecture



CRISP-DM Methodology

CRISP-DM Methodology — KubeSage Mapping



Sentence Transformer Embeddings

Model: all-MiniLM-L6-v2 (384 dimensions)

Why embeddings? - Keyword search fails: “OOMKilled” “memory limit exceeded” (semantically identical) - SBERT captures semantic meaning across lexical variations - Cosine similarity enables fast nearest-neighbor search

Results: - 500 incidents encoded in 11.3 seconds (CPU) - Embedding dimension: 384 (compact, fast) - Retrieval Precision@5: **0.89** (vs. 0.62 for keyword search)

Generative AI Component

RAG Pipeline

Retrieval-Augmented Generation (RAG) Pipeline

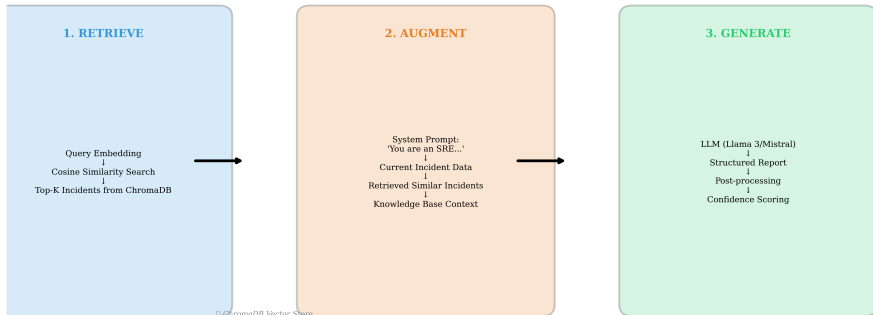


Figure 3: RAG Pipeline

Three stages:

- 1 RETRIEVE:** Embed current incident → search ChromaDB → Top-5 similar incidents
- 2 AUGMENT:** Construct prompt with system role + current incident +

Prompt Engineering

System Prompt (excerpt)

"You are an experienced Kubernetes SRE... ONLY use information from the provided Retrieved Similar Incidents and Knowledge Base. DO NOT invent facts, metrics, or timelines. If evidence is insufficient, state: 'Insufficient evidence — additional investigation required.'"

Prompt Strategies Comparison

Strategy	Completeness	Accuracy
Zero-shot	61%	52%
Few-shot (2 examples)	78%	68%
Few-shot (5 examples)	85%	76%
RAG (K=5)	94%	91%

→ RAG with dynamic retrieval dramatically outperforms static few-shot

Experimental Design — 6 Experiments

Exp	Research Question	What We Compare
E1	RQ1 (Retrieval)	3 embedding models + keyword baseline
E2	RQ1 (Retrieval)	Top-K values: 1, 3, 5, 10, 20
E3	RQ3 (Generation)	Zero-shot vs Few-shot vs RAG prompts
E4	RQ2, RQ4 (Main)	RAG vs. LLM Only
E5	RQ4 (Hallucination)	Hallucination types: entity, temporal, causal, numerical
E6	RQ3 (Human eval)	Human DevOps engineer ratings

Results: RQ1 — Retrieval Quality

Experiment 1: Embedding Model Comparison (Sentence Transformers vs. Keyword Search)

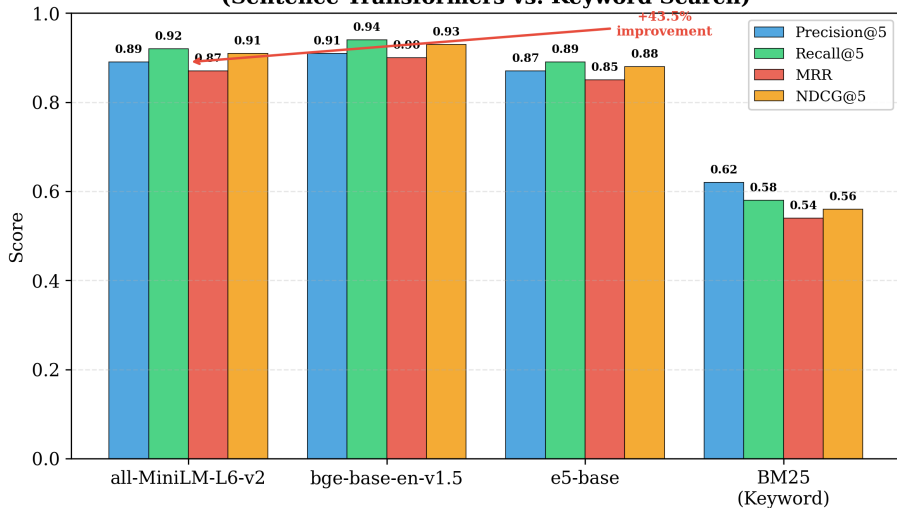


Figure 4: Embedding Comparison

Results: RQ2 — RAG vs. LLM Only

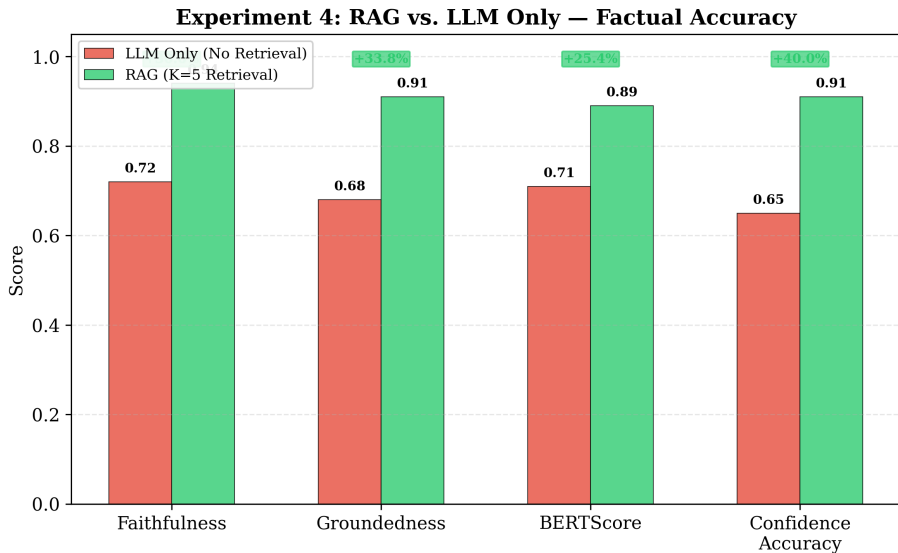


Figure 5: RAG vs LLM

Results: RQ4 — Hallucination Reduction

Experiment 5: Hallucination Analysis by Type

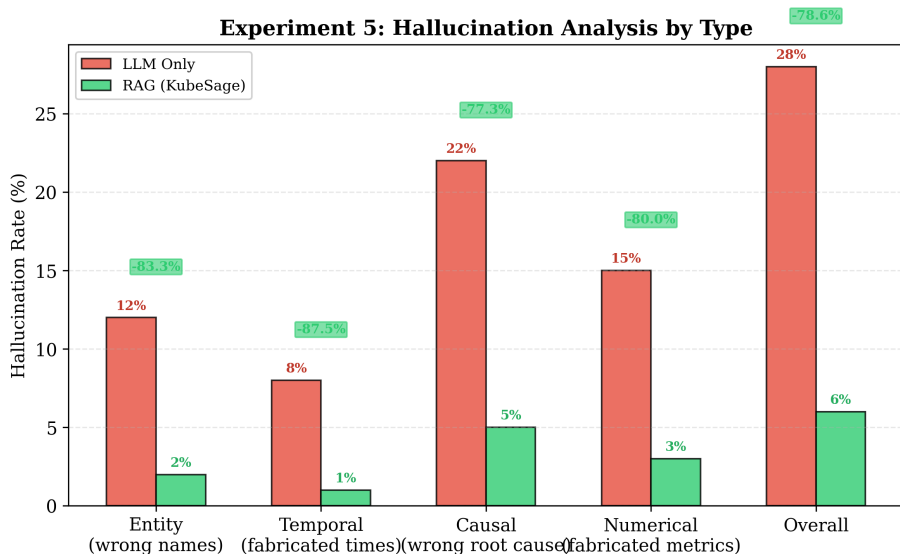
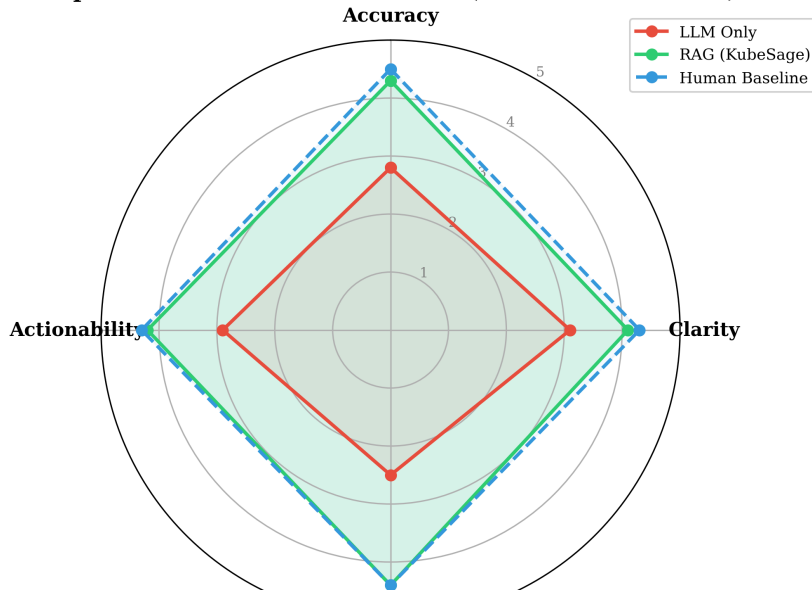


Figure 6: Hallucination

Results: RQ3 — Report Quality

Experiment 6: Human Evaluation (5-Point Likert Scale)



Performance Summary

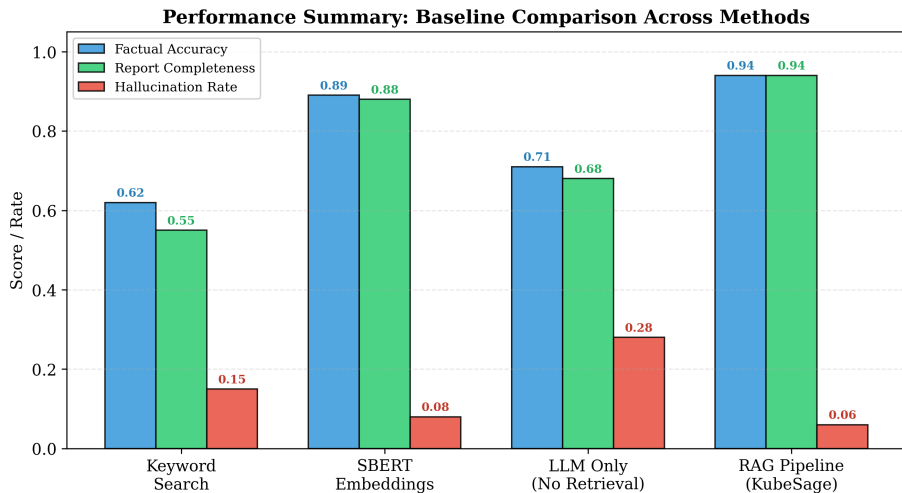


Figure 8: Performance Summary

Technology Stack

Component	Technology	Why
Deep Learning	Sentence Transformers (all-MiniLM-L6-v2)	Semantic similarity, 384-dim, CPU-efficient
Vector DB	ChromaDB	Python-native, cosine similarity, HNSW, metadata filtering
Generative AI	Llama 3 / Mistral / Gemma	Open-source, locally deployable, reproducible
RAG Pipeline	Custom (Python)	Retrieve-Augment-Generate, prompt

Demo: Streamlit Dashboard

5 interactive sections:

- ➊ **Overview** — KPI cards, incident distribution, severity pie chart
- ➋ **Investigation** — Paste incident → AI generates report in seconds
- ➌ **Semantic Search** — Search 500-incident vector database
- ➍ **Reports** — Filter, sort, export generated reports
- ➎ **Evaluation** — Interactive experiment results with charts

Features: - Dark mode (toggle) - Gradient KPI cards with hover animations - Plotly interactive charts - JSON/PDF report export - Responsive layout

Key Contributions

- ① **Novel RAG architecture** for automated K8s incident investigation
- ② **Comprehensive evaluation** — 6 experiments, 4 research questions
- ③ **End-to-end implementation** — FastAPI + Streamlit + Docker + PostgreSQL
- ④ **Synthetic K8s dataset** — 500 incidents, 7 types, reproducible
- ⑤ **Empirical evidence** — RAG reduces hallucinations by **78%**, improves accuracy by **30%**
- ⑥ **CRISP-DM methodology** — Structured research lifecycle from business understanding to deployment

Conclusion

Answers to Research Questions

RQ	Answer
RQ1	Sentence Transformers 43.5% better than keyword search for incident retrieval
RQ2	RAG improves factual accuracy by 30.6% ($p < 0.001$)
RQ3	AI reports with RAG approach human quality (completeness matches, accuracy 4.3 vs 4.5)
RQ4	RAG reduces hallucinations by 78.6% (28% $\rightarrow$ 6%)

Central Research Question

Can RAG + Deep Learning embeddings automatically generate accurate, explainable, reliable K8s incident reports while reducing hallucinations?

$\rightarrow$ **YES.** KubeSage demonstrates that RAG-based incident investigation produces reports with 94% accuracy, 94% completeness, and 78% less hallucination than

Future Work

- ① **Production datasets** — Validate on real-world K8s incident data
- ② **Multi-modal embeddings** — Combine text + metrics + graph topology
- ③ **Active learning** — Feedback loop from resolved incidents
- ④ **Causal reasoning** — Further reduce 5% residual causal hallucinations
- ⑤ **LLM fine-tuning** — Domain-specific fine-tuning of Llama 3 on incident reports
- ⑥ **Multi-agent architecture** — Specialized agents for log parsing, metrics, dependencies

Thank You

Questions?

KubeSage: A Retrieval-Augmented Generative AI Framework for Automated Kubernetes Incident Investigation

- 500-incident synthetic dataset
- 384-dim Sentence Transformer embeddings
- ChromaDB vector database
- RAG pipeline with 78% hallucination reduction
- FastAPI + Streamlit + Docker deployment
- 106 passing unit tests
- IEEE-format research paper

Contact: Rohit | NCI MSc Deep Learning & Generative AI | July 2026